

Classical Mechanics

Lecture 3: Stationary Action and the Euler–Lagrange Equation

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Chapter 1

Stationary Action and the Euler–Lagrange Equation

Overview. The familiar name *principle of least action* hides the essential point: the physical trajectory makes the action stationary, but need not minimize it. We shall build that statement from ordinary equilibrium, extend it from points to entire functions, derive the Euler–Lagrange equation by discretizing a path, and recover Newton’s equations. Rotating and polar coordinates will then show why the Lagrangian formulation is more than a change of notation.

1.1 Stationary, Not Necessarily Least

The word *least* is the unreliable part of the traditional name. A classical trajectory does not always make the action a minimum. It may instead give a maximum or a saddle in the space of possible trajectories. The correct principle is that the action is *stationary*: its first-order change vanishes.

Begin with the simpler question of equilibrium. A particle that remains at rest has no acceleration, so its net force must vanish. For a force derived from a one-dimensional potential,

$$F(x) = -\frac{dV}{dx}, \quad (1.1)$$

the equilibrium condition is

$$\frac{dV}{dx} = 0. \quad (1.2)$$

Now displace the particle by a small amount δx . To first order,

$$\delta V = V(x + \delta x) - V(x) = \frac{dV}{dx} \delta x + O(\delta x^2). \quad (1.3)$$

At equilibrium the linear term is zero for either sign of δx :

$$\delta V = 0 \quad \text{to first order.} \quad (1.4)$$

This does not say that V is exactly unchanged after a finite move. It says that the leading change is quadratic or higher. A minimum and a maximum both satisfy this condition.

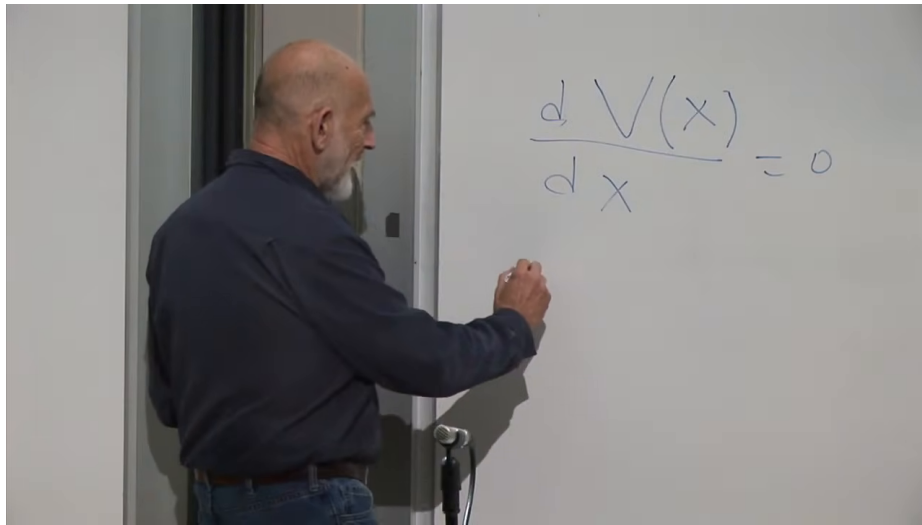


Figure 1.1: The one-dimensional equilibrium condition: force vanishes where the potential is stationary.

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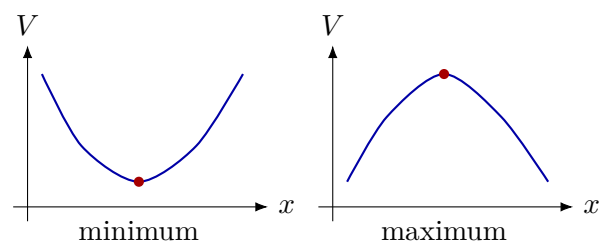


Figure 1.2: Stationarity fixes the first derivative, not whether the point is a minimum or a maximum.

For many coordinates q_i , the first variation is

$$\delta V = \sum_i \frac{\partial V}{\partial q_i} \delta q_i. \quad (1.5)$$

If this must vanish for every independent infinitesimal displacement δq_i , then every coefficient vanishes:

$$\frac{\partial V}{\partial q_i} = 0 \quad \text{for every } i. \quad (1.6)$$

A two-dimensional potential is better imagined as hilly terrain than as a roller coaster. A stationary point may be a valley bottom, a hilltop, or a saddle. At a saddle, motion in one direction initially climbs while motion in another initially descends. The restoring behavior can therefore be stable in one direction and unstable in another, even though all first derivatives vanish at the point itself.

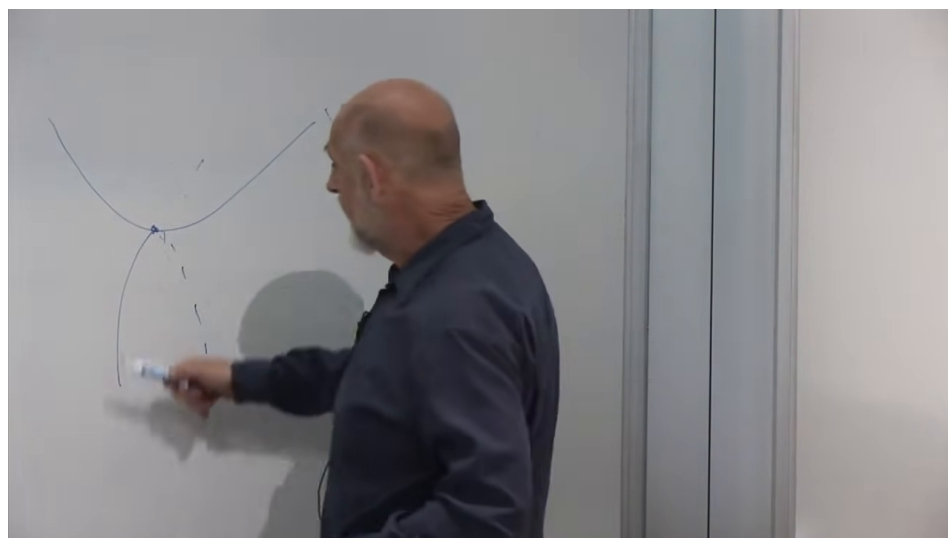


Figure 1.3: A saddle is stationary in every first-order direction, although neighboring displacements rise along one axis and fall along another.

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In a space with many coordinates, saddle points are more typical than pure maxima or minima. This is why *stationary* is the durable word. The condition

$$\delta V = 0 \quad (1.7)$$

summarizes equilibrium without prejudging the second-order behavior.

1.2 From Points to Functions

Mechanics concerns whole motions, not isolated equilibrium points. A trajectory is a function such as $x(t)$, or a collection $(x(t), y(t), z(t))$. The mathematical problem of making a quantity stationary by varying an entire function is the *calculus of variations*. Before applying it to motion, consider three simpler examples.

1.2.1 Shortest curves and geodesics

Fix two points in the plane and represent a curve between them by $y(x)$. Over a short segment,

$$ds = \sqrt{dx^2 + dy^2} = dx \sqrt{1 + \left(\frac{dy}{dx}\right)^2}, \quad (1.8)$$

where the last form assumes $dx > 0$. The total length is the functional

$$S[y] = \int_{x_0}^{x_1} \sqrt{1 + y'(x)^2} dx. \quad (1.9)$$

Square brackets emphasize that S takes an entire function as input. The straight line is selected by

$$\delta S = 0 \quad (1.10)$$

under small changes of the curve that leave both endpoints fixed.

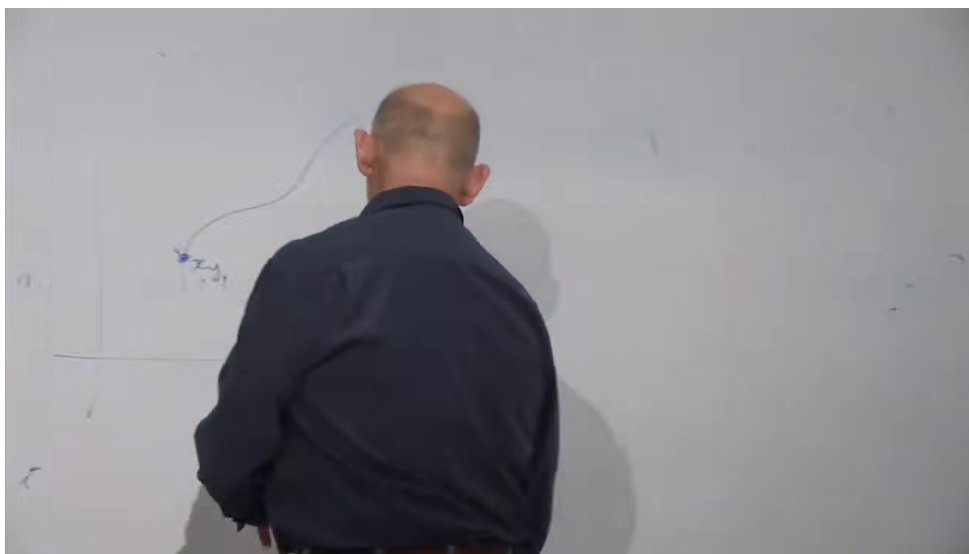


Figure 1.4: A trial curve between two fixed endpoints. Its length is the sum of the infinitesimal hypotenuses ds .

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The same question can be asked on a curved surface. Pulling a string tight between two stakes gives a physical picture of the shortest constrained route. Such a stationary-length curve is a *geodesic*. On a flat plane it is a straight line; on a sphere or hilly surface it generally is not.

1.2.2 Fermat's principle

A light ray in a medium with position-dependent speed $c(x, y)$ need not follow the shortest spatial route. The time required for a small segment is distance divided by local speed, so

$$T[y] = \int_{x_0}^{x_1} \frac{\sqrt{1 + y'(x)^2}}{c(x, y)} dx. \quad (1.11)$$

Fermat's principle requires

$$\delta T = 0. \quad (1.12)$$

When c is constant this reduces to the shortest-distance problem. In a variable refractive medium, however, the fastest path balances distance against local propagation speed.

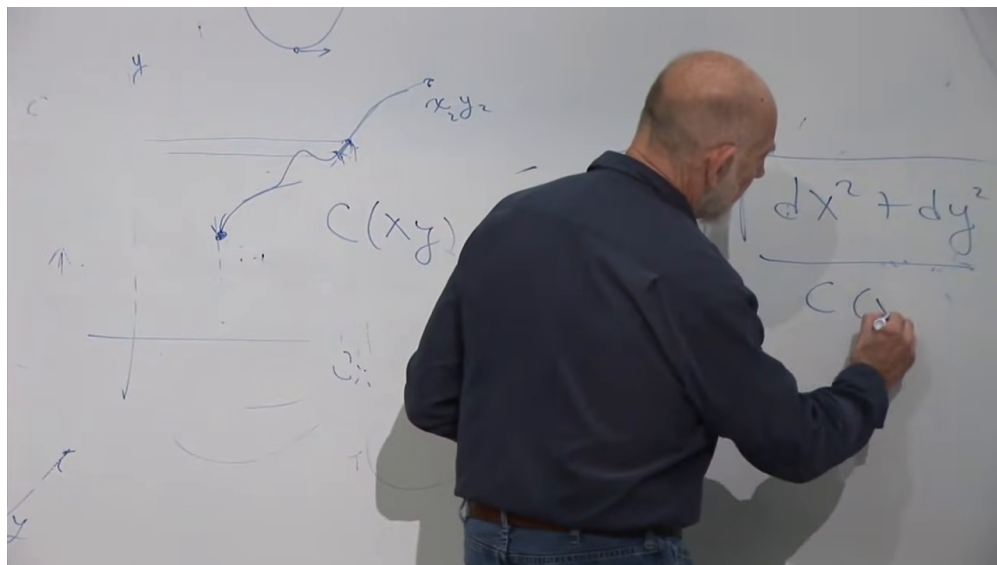


Figure 1.5: The least-time functional for a ray in a medium whose local light speed depends on position.

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A topical joke invokes the then-recent claim that neutrinos moved unusually fast in Italy. The physics argument immediately returns to the reliable example: ordinary light changes speed in material according to the index of refraction.

1.2.3 A hanging chain

A chain pinned at two points settles into a catenary. Each small piece contributes gravitational potential energy, so the total potential is itself an integral over the chain. Once again the endpoints are fixed, the unknown shape is a function, and an integral must be stationary.

These examples have the same architecture: two fixed endpoints, a function between them, and an integral whose first variation vanishes. Reducing such a global instruction to a differential equation is what it means to solve the variational problem.

1.3 Mechanics as a Boundary-Value Problem

A trajectory is often drawn with time horizontal because t is the independent variable. There is nevertheless a strong convention in physics to draw time upward. An aside recalls experiments in which people ordered pictures from youth to age in directions associated with their writing systems: left-to-right readers tended to arrange them left to right, right-to-left readers the opposite way, and vertically trained readers vertically. Physicists, the joke concludes, are the people who insist that

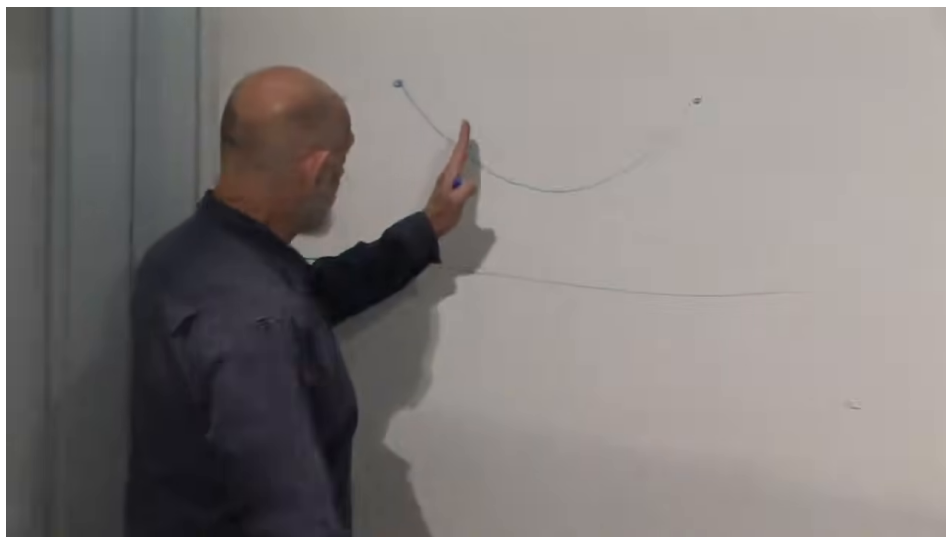


Figure 1.6: A hanging chain between fixed supports. Its stationary shape is selected by the total gravitational potential energy.

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time flows upward. We shall use the conventional horizontal t -axis here because it makes $x(t)$ read naturally as a graph.

In the initial-value formulation, the data at t_0 are

$$x(t_0) = x_0, \quad \dot{x}(t_0) = v_0. \quad (1.13)$$

Newton's equation then determines the subsequent orbit. A variational formulation poses a different problem: specify two endpoint events,

$$x(t_0) = x_0, \quad x(t_1) = x_1, \quad (1.14)$$

and determine the physical path between them. The initial velocity is no longer supplied; it becomes part of the solution.

The quantity to be made stationary is the *action*,

$$A[x] = \int_{t_0}^{t_1} L(x, \dot{x}, t) dt. \quad (1.15)$$

The integrand L is the *Lagrangian*. It may depend on position, velocity, and explicitly on time. The action principle is

$$\delta A = 0 \quad (1.16)$$

for all small path variations with

$$\delta x(t_0) = \delta x(t_1) = 0. \quad (1.17)$$

The endpoint condition is essential: we compare neighboring paths that connect the same two events.

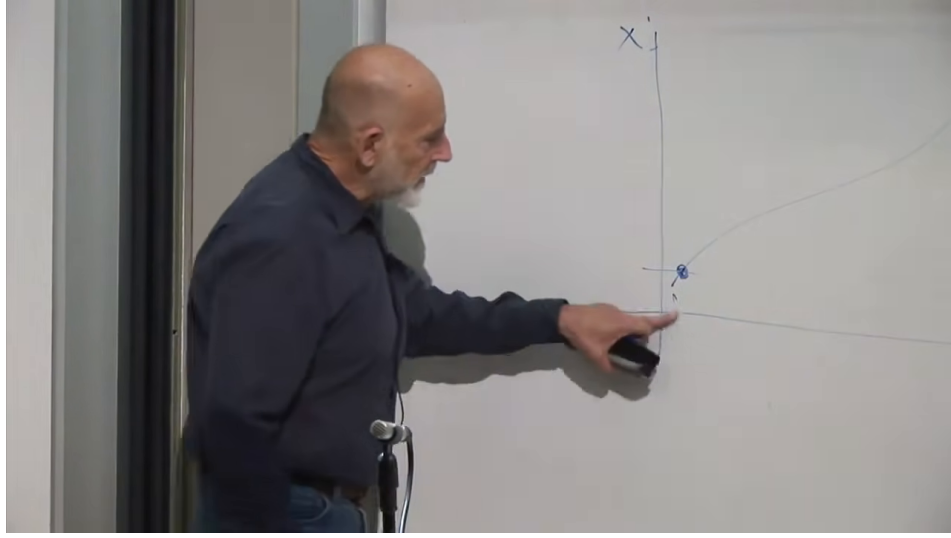


Figure 1.7: A mechanical trajectory $x(t)$. One may prescribe position and slope at one end, or positions at two endpoint times.

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1.4 A Discrete Derivation of the Euler–Lagrange Equation

The action is a functional of infinitely many path values. To make its variation concrete, divide the time interval into steps of length ϵ ,

$$t_n = t_0 + n\epsilon, \quad x_n = x(t_n), \quad (1.18)$$

and approximate the velocity on the n -th interval by

$$v_n = \frac{x_{n+1} - x_n}{\epsilon}. \quad (1.19)$$

The integral becomes the ordinary sum

$$A_\epsilon = \sum_{n=0}^{N-1} \epsilon L \left(x_n, \frac{x_{n+1} - x_n}{\epsilon} \right). \quad (1.20)$$

We use n for the time-slice label so that i remains available later for spatial coordinates and particles.

Vary one interior point x_n , holding every other x_k fixed. Only two terms in the sum depend on x_n :

$$\begin{aligned} A_\epsilon|_{x_n} &= \epsilon L \left(x_n, \frac{x_{n+1} - x_n}{\epsilon} \right) \\ &\quad + \epsilon L \left(x_{n-1}, \frac{x_n - x_{n-1}}{\epsilon} \right). \end{aligned} \quad (1.21)$$

Differentiating gives

$$0 = \frac{\partial A_\epsilon}{\partial x_n} = \epsilon \frac{\partial L_n}{\partial x_n} - \frac{\partial L_n}{\partial v_n} + \frac{\partial L_{n-1}}{\partial v_{n-1}}. \quad (1.22)$$

The opposite signs have a simple geometric origin. Raising x_n increases the slope of the interval behind it and decreases the slope of the interval ahead of it.

$$\int dt \rightarrow \epsilon \sum_i \mathcal{L}\left(x_i, \frac{x_{i+1} - x_i}{\epsilon}\right)$$

Figure 1.8: The trajectory replaced by a chain of time slices. Varying one interior height changes only its two neighboring segments.

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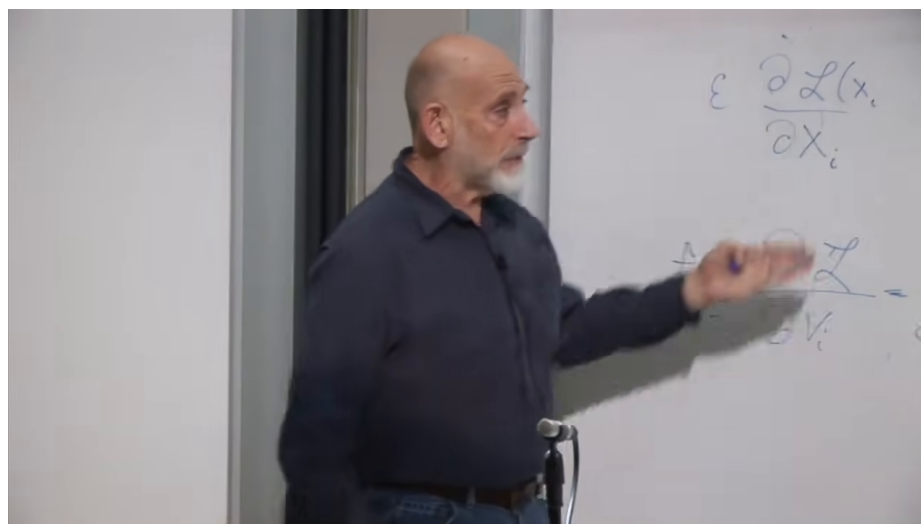


Figure 1.9: The direct position derivative and the two neighboring velocity-derivative contributions in the discrete variation.

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Rearrange Eq. (1.22):

$$\frac{1}{\epsilon} \left(\frac{\partial L_n}{\partial v_n} - \frac{\partial L_{n-1}}{\partial v_{n-1}} \right) = \frac{\partial L_n}{\partial x_n}. \quad (1.23)$$

The left side is a neighboring difference quotient. As $\epsilon \rightarrow 0$, the broken line becomes a smooth path and the quotient becomes a time derivative:

$$\boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) = \frac{\partial L}{\partial x}}. \quad (1.24)$$

This is the Euler–Lagrange equation.

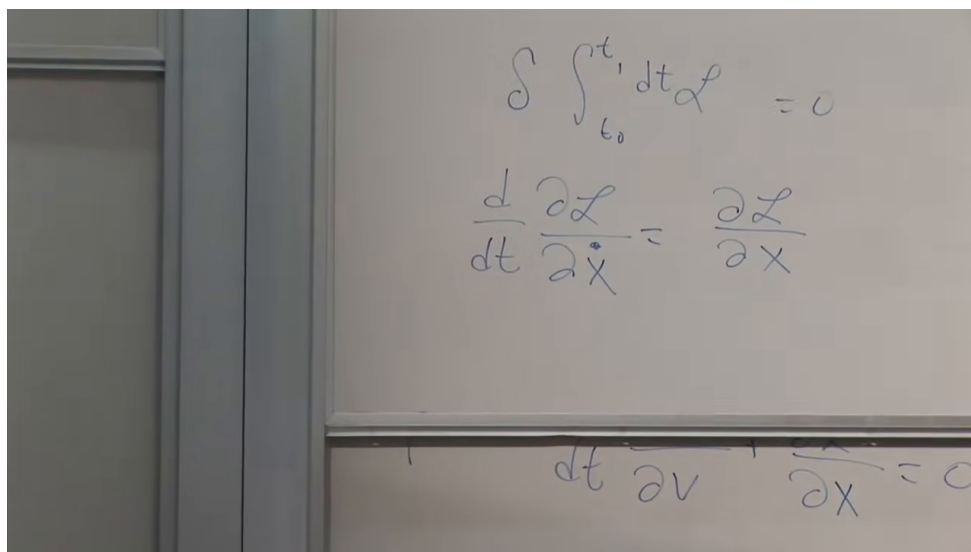


Figure 1.10: The completed Euler–Lagrange equation obtained from stationarity at every interior time slice.

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Once L is specified, Eq. (1.24) acts as a machine: differentiate the given function, and the machine returns a differential equation for the path.

Question from the audience. Is the Euler–Lagrange result valid for any function of x and $\dot{x}$?^a

Response. Yes, subject to the regularity needed for the derivatives. The arc-length example has exactly the same form after relabeling its independent variable x as the role played by t , and its unknown function $y(x)$ as the role played by $x(t)$. Its Euler–Lagrange equation has straight-line solutions.

^aLecture timestamp: 00:49:37–00:50:24.

1.5 Newton’s Equations from the Lagrangian

For a particle on a line in a potential $V(x)$, take

$$L(x, \dot{x}) = T - V = \frac{1}{2} m \dot{x}^2 - V(x). \quad (1.25)$$

The minus sign may look strange because total energy is $T + V$. But the Lagrangian is not the energy. Its test is whether the Euler–Lagrange machine returns the correct motion. A mock vote about the sign is declined: the answer must be checked, not elected.

First,

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}. \quad (1.26)$$

This is momentum. More generally, the momentum conjugate to any coordinate q is defined by

$$p_q = \frac{\partial L}{\partial \dot{q}}. \quad (1.27)$$

Taking the time derivative of Eq. (1.26) gives $m\ddot{x}$, while

$$\frac{\partial L}{\partial x} = -\frac{dV}{dx}. \quad (1.28)$$

The Euler–Lagrange equation is therefore

$$m\ddot{x} = -\frac{dV}{dx} = F(x), \quad (1.29)$$

which is Newton’s second law.

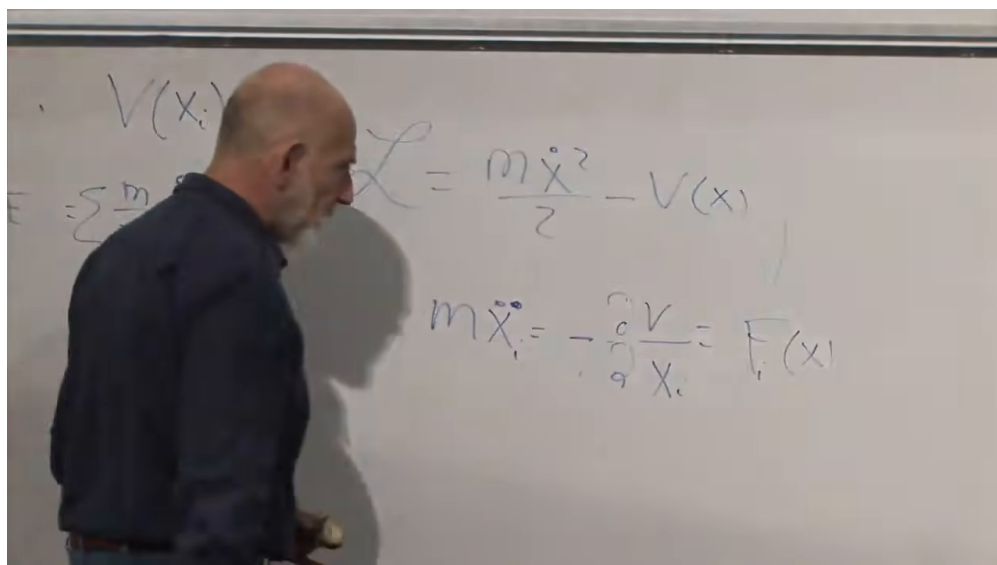


Figure 1.11: Kinetic energy, potential energy, $L = T - V$, and the recovered component form of Newton’s equation on one board.

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For many coordinates q_i ,

$$L(\mathbf{q}, \dot{\mathbf{q}}) = \sum_i \frac{1}{2} m_i \dot{q}_i^2 - V(q_1, q_2, \dots), \quad (1.30)$$

and there is one equation for each coordinate:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i} \implies m_i \ddot{q}_i = -\frac{\partial V}{\partial q_i} = F_i. \quad (1.31)$$

The trajectory is still one curve, now through a high-dimensional configuration space.

Question from the audience. Have we merely worked backward from Newton's law and engineered a functional that reproduces the answer?^a

Response. That is not the productive viewpoint. The Lagrangian is the compact specification of the physical system: one function of all coordinates and velocities packages all its equations of motion. In new theories, physicists usually propose an action and derive the equations, rather than guessing every equation separately.

^aLecture timestamp: 00:58:07–01:00:04.

This packaging is one advantage. Coordinate freedom is another. The statement that a line is shortest does not depend on whether the plane is described by Cartesian, polar, or curvilinear coordinates. In the same way, stationarity of the action is coordinate-independent, even though the formula for L and the resulting component equations change.

1.6 A Free Particle Seen from a Carousel

Let (x, y) be inertial Cartesian coordinates and let (X, Y) rotate uniformly with angular velocity ω . With the orientation used here,

$$\begin{aligned} x &= X \cos \omega t + Y \sin \omega t, \\ y &= -X \sin \omega t + Y \cos \omega t. \end{aligned} \tag{1.32}$$

The angle between the frames is ωt .

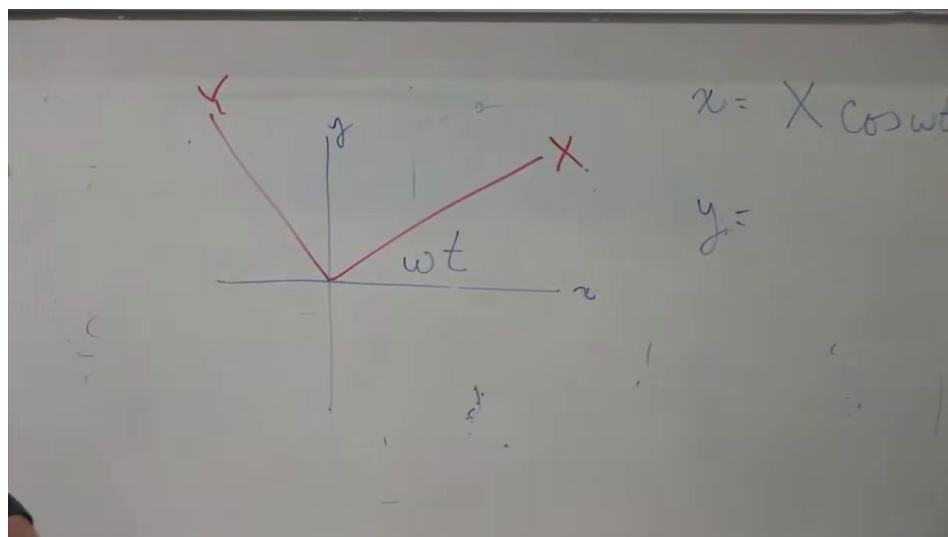


Figure 1.12: Inertial and uniformly rotating Cartesian axes. The same point receives time-dependent rotating coordinates.

Lecture frame: 01:06:05.

A free particle has inertial-frame Lagrangian

$$L_{\text{in}} = \frac{m}{2} (\dot{x}^2 + \dot{y}^2). \tag{1.33}$$

Differentiate Eq. (1.32) and use $\sin^2 \omega t + \cos^2 \omega t = 1$. The trigonometric terms combine to give

$$\dot{x}^2 + \dot{y}^2 = \dot{X}^2 + \dot{Y}^2 + \omega^2(X^2 + Y^2) + 2\omega(Y\dot{X} - X\dot{Y}). \quad (1.34)$$

Thus

$$L_{\text{rot}} = \frac{m}{2}(\dot{X}^2 + \dot{Y}^2) + \frac{m\omega^2}{2}(X^2 + Y^2) + m\omega(Y\dot{X} - X\dot{Y}). \quad (1.35)$$

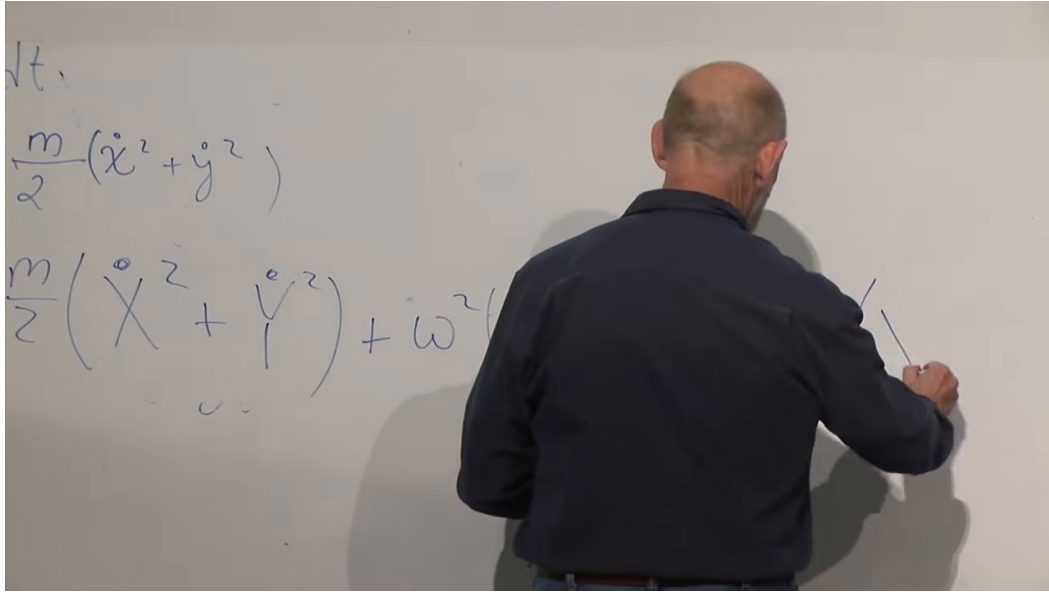


Figure 1.13: The transformed free-particle Lagrangian: ordinary kinetic energy, a quadratic centrifugal term, and a term linear in velocity.

Lecture frame: 01:12:16.

The quadratic coordinate term has the form $-V_{\text{fict}}$, where

$$V_{\text{fict}} = -\frac{m\omega^2}{2}(X^2 + Y^2). \quad (1.36)$$

It decreases outward, so it produces the centrifugal force $m\omega^2(X, Y)$.

The final term in Eq. (1.35) is linear in both coordinates and velocities. Applying the Euler–Lagrange equation gives

$$\begin{aligned} m\ddot{X} &= m\omega^2 X - 2m\omega\dot{Y}, \\ m\ddot{Y} &= m\omega^2 Y + 2m\omega\dot{X}. \end{aligned} \quad (1.37)$$

The terms proportional to ω^2 are centrifugal; those proportional to velocity are Coriolis. In vector form within the rotating plane,

$$\mathbf{F}_C = -2m\boldsymbol{\omega} \times \mathbf{v}_{\text{rot}}, \quad (1.38)$$

with signs adjusted if the opposite orientation convention is chosen.

The point is not that the algebra disappears. It is that one coordinate transformation of a single scalar function generates both coupled equations and both fictitious forces.

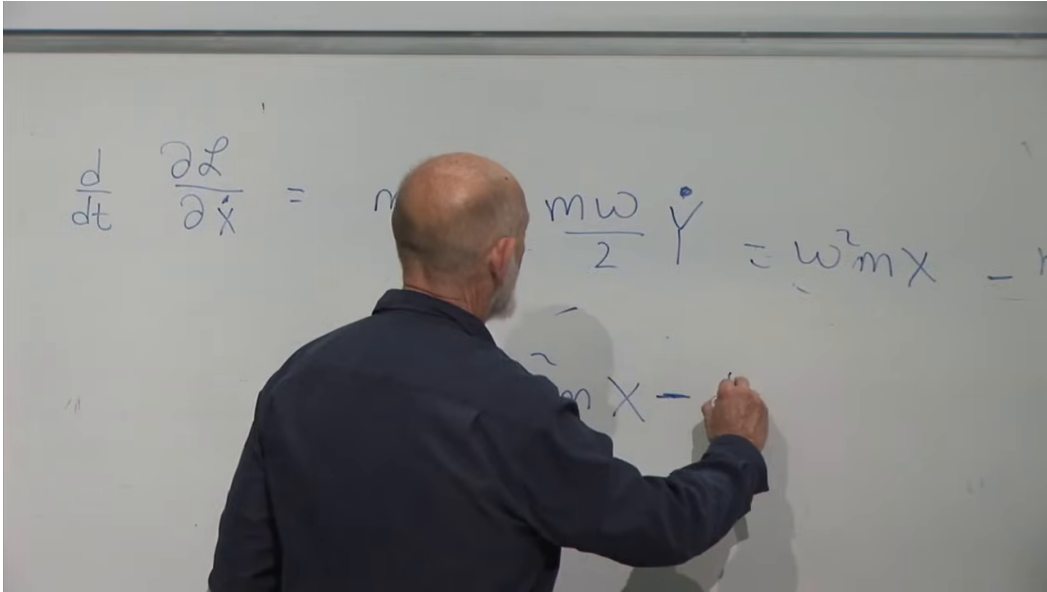


Figure 1.14: The rotating-frame equations. Coordinate-proportional terms give centrifugal acceleration; cross-velocity terms give Coriolis acceleration.

Lecture frame: 01:20:20.

Question from the audience. Does this method work for arbitrarily curved coordinates?^a

Response. Yes. The new coordinates may be curvilinear and may even depend explicitly on time, as the rotating coordinates do. Rewrite the action in the new variables and apply the same Euler–Lagrange rule.

^aLecture timestamp: 01:22:01–01:22:17.

1.7 Polar Coordinates and a Cyclic Angle

For ordinary polar coordinates,

$$x = r \cos \theta, \quad y = r \sin \theta. \quad (1.39)$$

Both r and θ may vary with time. Differentiating and adding the squares yields

$$\dot{x}^2 + \dot{y}^2 = \dot{r}^2 + r^2 \dot{\theta}^2. \quad (1.40)$$

For a central potential $V(r)$,

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r). \quad (1.41)$$

The radial equation is

$$m\ddot{r} = mr\dot{\theta}^2 - \frac{dV}{dr}. \quad (1.42)$$

The first term is the outward centrifugal contribution associated with the particle's own angular motion.

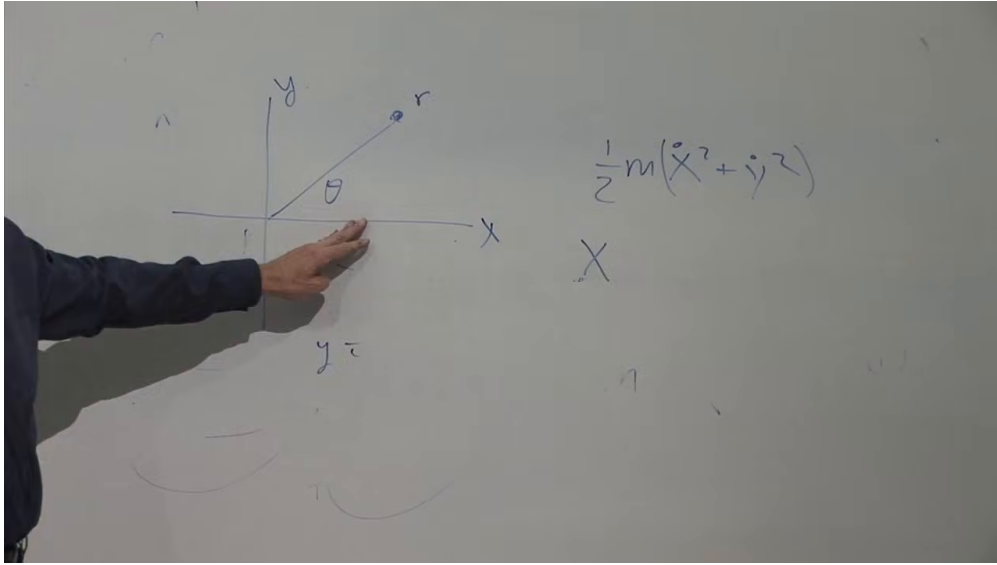


Figure 1.15: Polar geometry and the decomposition of kinetic energy into radial and angular motion.

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For the angle,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = \frac{\partial L}{\partial \theta} = 0, \quad (1.43)$$

because the central potential has no explicit θ -dependence. Hence

$$\frac{d}{dt} (mr^2\dot{\theta}) = 0, \quad mr^2\dot{\theta} = \ell, \quad (1.44)$$

where ℓ is the conserved angular momentum.

The angular equation gives

$$\dot{\theta} = \frac{\ell}{mr^2}. \quad (1.45)$$

Substitution into Eq. (1.42) leaves a single equation for r :

$$m\ddot{r} = \frac{\ell^2}{mr^3} - \frac{dV}{dr}. \quad (1.46)$$

A coordinate absent from L is traditionally called *cyclic*, even when it is not an angle. If q is cyclic, then

$$\frac{\partial L}{\partial q} = 0 \implies \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = 0. \quad (1.47)$$

Its conjugate momentum is conserved. This is the first glimpse of a general link between symmetry and conservation law.

1.8 Questions About Meaning, Energy, and Scope

The questions that follow expose where the deeper content of the formalism lies.

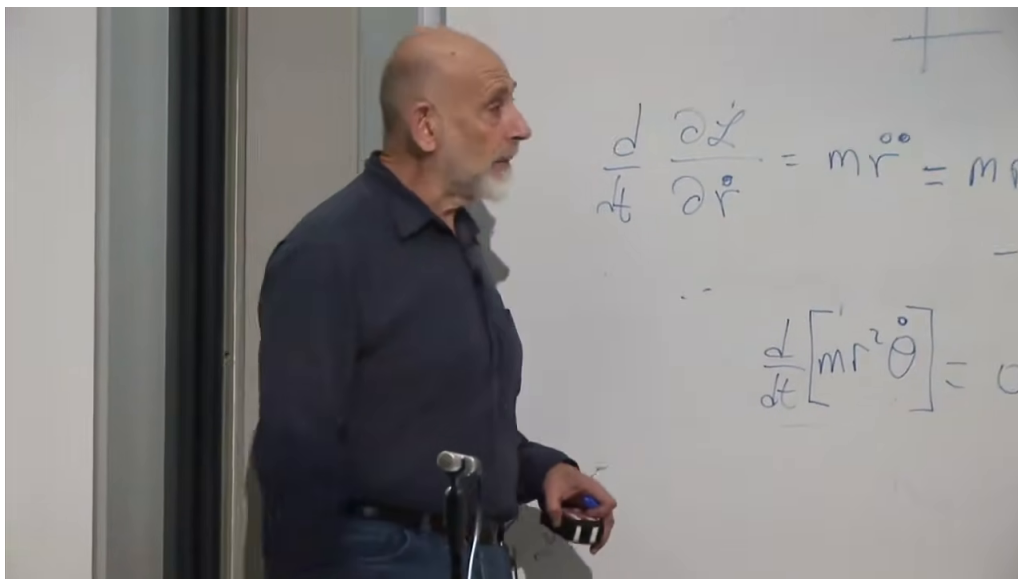


Figure 1.16: The angular Euler–Lagrange equation has zero right-hand side, exposing conservation of $mr^2\dot{\theta}$.

Lecture frame: 01:29:25.

Question from the audience. Is there an independent reason to arrive at the Lagrangian, beyond discovering that it gives the right classical equations?^a

Response. Historically, more than a century of classical mechanics showed that particles, waves, and electromagnetic fields could all be cast in this form. At the classical level that accumulated success is an empirical fact about how laws are organized. The deeper explanation comes from quantum mechanics and is deferred.

^aLecture timestamp: 01:33:25–01:35:07.

Question from the audience. In the rotating-frame Lagrangian, can the velocity-linear term be assigned an ordinary kinetic or potential energy?^a

Response. No. The conserved quantity associated with time translation is obtained by the Hamiltonian construction, not by reading every term in L as kinetic or potential energy. For Eq. (1.35), the velocity-linear term cancels from $\sum_a \dot{q}_a \partial L / \partial \dot{q}_a - L$. The Hamiltonian treatment will organize this systematically.

^aLecture timestamp: 01:35:08–01:36:23.

Indeed, because L_{rot} has no explicit time dependence,

$$H_{\text{rot}} = \dot{X} \frac{\partial L}{\partial \dot{X}} + \dot{Y} \frac{\partial L}{\partial \dot{Y}} - L = \frac{m}{2} (\dot{X}^2 + \dot{Y}^2) - \frac{m\omega^2}{2} (X^2 + Y^2) \quad (1.48)$$

is conserved. It is the rotating-frame Jacobi integral, not the same numerical energy assigned by the inertial observer. Time-dependent coordinate changes need not preserve the numerical form of energy.

Question from the audience. What if a curved trajectory is caused by gravity?^a

Response. For Newtonian gravity, simply use $V(r) = -GMm/r$ in the central-force Lagrangian. Curved spacetime belongs to general relativity and is not assumed here.

^aLecture timestamp: 01:36:30–01:37:10.

Question from the audience. Are the centrifugal potential and its energy merely fictitious?^a

Response. An observer on the rotating platform measures the effective forces and can consistently use the rotating-frame conserved quantity. *Fictitious* means that these terms arise from a noninertial description, not that their observed effects are imaginary. Energy itself is not invariant under a general time-dependent change of coordinates.

^aLecture timestamp: 01:37:13–01:40:14.

Question from the audience. What direction does the velocity-dependent Coriolis force point?^a

Response. It is perpendicular to the rotating-frame velocity, with magnitude proportional to speed and to the frame's angular velocity. In the plane the components exchange $\dot{X}$ and $\dot{Y}$ with opposite signs, as in Eq. (1.37).

^aLecture timestamp: 01:40:22–01:42:36.

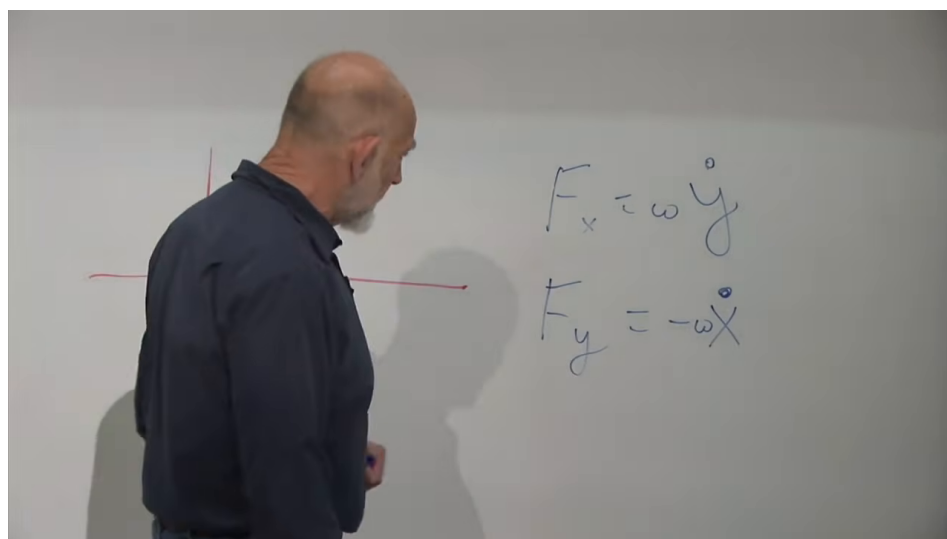


Figure 1.17: The cross-velocity component pattern: the Coriolis force is perpendicular to the rotating-frame velocity.

Lecture frame: 01:41:40.

Question from the audience. Must the starting Lagrangian be written in an inertial frame?^a

Response. For an ordinary free particle, Newton's inertial-frame law supplies the simple starting form $m(\dot{x}^2 + \dot{y}^2)/2$. Once that physical law is known, the action can be transformed to any other coordinates. One should not assume the same simple kinetic formula independently in a rotating frame.

^aLecture timestamp: 01:42:37–01:44:00.

The same type of velocity-linear term occurs for a charged particle in a magnetic field:

$$L_{\text{em}} = \frac{1}{2}m\mathbf{v}^2 + q\mathbf{A}(\mathbf{r}, t) \cdot \mathbf{v} - q\phi(\mathbf{r}, t). \quad (1.49)$$

For a uniform magnetic field perpendicular to the plane, the symmetric gauge $\mathbf{A} = \frac{1}{2}\mathbf{B} \times \mathbf{r}$ produces a term proportional to $X\dot{Y} - Y\dot{X}$, the same algebraic structure as the Coriolis term.

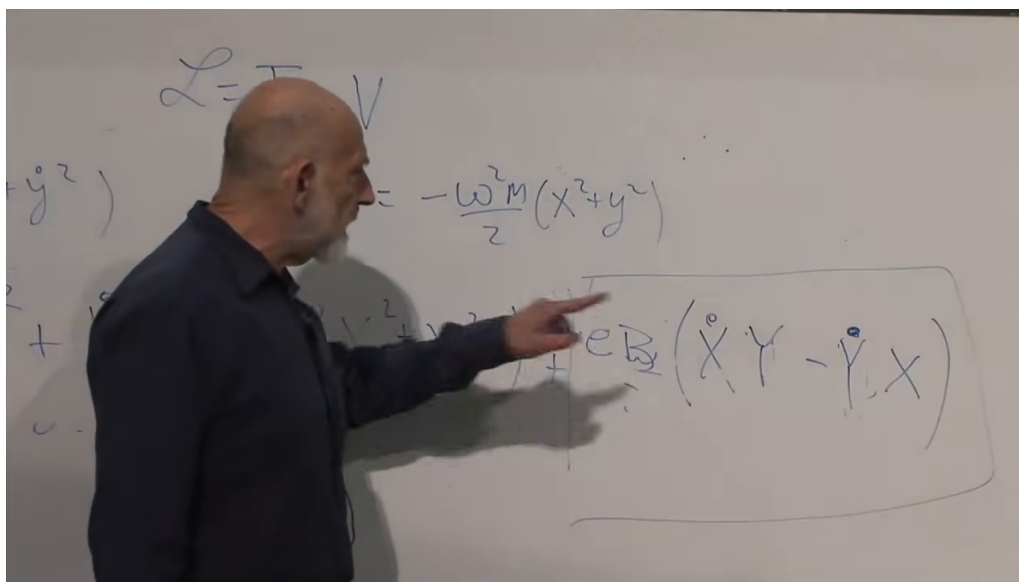


Figure 1.18: The velocity-linear rotating-frame term that motivates the comparison with a charged particle in a uniform magnetic field.

Lecture frame: 01:44:40.

Question from the audience. Are there constraints on what function may serve as a Lagrangian, beyond smoothness?^a

Response. Yes, but the relevant constraints require more structure than has yet been developed and are deferred. At this stage the central lesson is how a proposed Lagrangian generates equations and conservation laws.

^aLecture timestamp: 01:44:46–01:45:06.

The rotating equations also admit circular motion: a point fixed in the inertial frame is seen from the carousel to circle the rotating origin. The magnetic analogy then explains why a charged particle can execute circular cyclotron motion.

Question from the audience. Would a charged particle remain on that circle, or would radiation make the orbit collapse?^a

Response. The particle-only equations being discussed omit radiation. Once coupling to the electromagnetic field and radiation reaction are included, an accelerating classical charge can radiate. Radiation is therefore a further physical effect, not a contradiction of the orbit in the truncated equations.

^aLecture timestamp: 01:46:11–01:47:21.

Question from the audience. Where do hurricanes enter this discussion? ^a

Response. Their large-scale atmospheric motion occurs on the rotating Earth, so Coriolis effects are important. A proper derivation belongs to meteorology and is not attempted here.

^aLecture timestamp: 01:47:30–01:47:50.

Question from the audience. Is the action framework expected to cover every physical law, and what would a genuine exception mean?^a

Response. Every known closed fundamental theory in the discussion's scope can be organized by an action: Newtonian particles, Maxwell fields, general relativity, and the Standard Model. Apparent counterexamples such as viscous drag arise when environmental degrees of freedom are omitted. Enlarge the system to include the fluid or microscopic bath and the fundamental description again fits the framework. A truly fundamental exception would therefore be a major discovery.

^aLecture timestamp: 01:47:57–01:49:01.